MPI LANCHESTER–OSIPOV

Methodology for assessing military power: criticism of models and rationale for improvements

An in-depth analysis of the Lanchester–Osipov model, a review of 13 existing approaches and 13 improvements built into the programme

Document 1 of 5 · Methodology

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Summary

The document substantiates the methodology of the programme “MPI Lanchester–Osipov” — a military power index (MPI) of states by 253 indicators in 12 pillars with a dynamic battle model. It answers three questions: what the classical Lanchester–Osipov model gives, why existing military power ratings are systematically wrong, and which specific improvements remove these errors.

Main conclusions

1. **Lanchester's model does not forecast war, but it sets the right mathematical form of the index.** Combat power grows as “quality × numbers to the power α”. In logarithms this is the weight rule `ln F = ln Q + α·ln N`, and it is precisely this that we build into the calculation of the combat pillars.
2. **The exponent α must be calibrated.** Fitting to the Ardennes, Kursk and historical battle databases does not confirm a pure square law (α = 2). The result depends on the level of combat, the type of weapon and the method of counting forces. Across 404 battles of 1900–1973 (CDB90 database), with equal combat effectiveness of the sides α ≈ 1.0 [0.8; 1.2]: since 13.10.2026 α = 1.0 by default, in the uncertainty calculation — 0.8–1.2 (document 6, section 7.5).
3. **Existing ratings count “items”.** Global Firepower, CINC and most indices linearly add up the quantity of equipment and resources with opaque or arbitrary weights. They do not account for quality, serviceability, the synergy of branches of service, geography, time, allies or uncertainty. The wars of 2022–2026 have shown what errors this leads to.
4. **The proposed methodology removes the errors systematically.** Each indicator has been assigned a role in the model. The combat pillars are calculated by Lanchester's law. The pillars are aggregated non-linearly (CES), nuclear deterrence is placed on a separate axis, power projection and resilience to a protracted war are taken into account, and allies are set by explicit coalitions with reliability. The Monte Carlo method gives an index interval and a range of ranks, while the weights are calibrated against historical outcomes.
5. **The index and the battle model are connected.** The structural index automatically supplies the inputs of the combat equations: initial potential, effectiveness, reserve, replacement rate and projection radius. Therefore a single data set answers both the question “who is stronger in general” and the question “what will happen in a specific theatre”.

> **Limitation.** the methodology assesses military potential, not the intentions of the sides and not the outcome of a particular war. All conclusions are published with uncertainty intervals and the methodology version.

1. What military power is and how to measure it

Military power is the ability of a state to apply organised violence to achieve political goals against a specific adversary, in a specific theatre and over a certain period of time. Four requirements for any methodology follow from this definition.

- **Relativity.** Power does not exist in a vacuum: it is always directed “against whom” and “where”. Comparison of pairs (dyads) and allowance for theatre distance are therefore needed.
- **Multidimensionality.** Ready forces (capacity, capability, readiness) and the ability to replace them (economy, industry, mobilisation) are different dimensions. They cannot be reduced to a single “sum of tanks”.
- **Time.** In a short war ready forces decide; in a protracted one, production and reserves. The methodology must allow the horizon to be changed.
- **Uncertainty.** Data on quality, serviceability and stocks are closed. An honest index shows a range, not a point.

Hence the triad borrowed from the Heritage Foundation but taken down to numbers: **capacity** (how much force), **capability** (how modern and effective it is), **readiness** (what share is combat-capable now). To it we add **sustainment** (how quickly losses are replaced) and **deterrence** (the nuclear threshold).

2. The Lanchester–Osipov model: an in-depth analysis

2.1. Origins

The model was proposed independently by two authors. **M. P. Osipov** published in 1915 in “Military Collection” the article “The influence of the numbers of the fighting sides on their losses”. It already contains both the linear and the quadratic dependencies, as well as a comparison with data from 19th-century battles. **F. W. Lanchester** set out the same idea in articles of 1914 and the book “Aircraft in Warfare: The Dawn of the Fourth Arm” (1916). In the Russian tradition the name “Lanchester–Osipov model” has been adopted, and the programme bears this name.

2.2. The family of laws

Notation: x, y are the numbers of the sides; a is the rate at which one unit of y disables units of x; b is the same for x.

Law	Equations	Invariant and derivation	Where applicable
Linear, area fire	dx/dt = −a·x·y dy/dt = −b·x·y	$b \cdot x_0$ vs. $a \cdot y_0$: power ∝ e·N	unaimed fire, area artillery
Quadratic	dx/dt = −a·y dy/dt = −b·x	$b \cdot x^2 - a \cdot y^2$ = const: power ∝ $e \cdot N^2$	aimed fire, precision-guided weapons
Logarithmic (Peterson)	dx/dt = −a·x	losses ∝ own numbers	non-combat losses, target saturation
Mixed (Deitchman, 1962)	dx/dt = −a·y dy/dt = −b·x·y	parabolic law	insurgents against a regular army
Generalised (Helmbold, 1965)	dx/dt = −a·x^(1−W)·y^W	b·x^(2W) − a·y^(2W) = const; α = 2W	calibration against history
With reinforcements	dx/dt = −a·y + P(t)	—	mobilisation, rotation, disengagement
Heterogeneous	$dx_i/dt = -\Sigma_j a_{ij} \cdot \varphi_{ji} \cdot y_j$	weights of the models — the eigenvector of the matrix	mixed force groupings
Stochastic	Markov chain	win probability instead of a point estimate	small numbers, uncertainty
Salvo model (Hughes, 1995)	$\Delta B = (\alpha \cdot A - b_3 \cdot B) / b_1$	discrete salvoes, interception, survivability	naval battle, missile strikes, air defence

Solution of the square law: $x(t) = x_0 \cdot ch(\sqrt{(ab)} \cdot t) - y_0 \cdot \sqrt{(a/b)} \cdot sh(\sqrt{(ab)} \cdot t)$. The winner is determined by the sign of the invariant $b \cdot x_0^2 - a \cdot y_0^2$. If x wins, it is left with $x_fin = \sqrt{(x_0^2 - (a/b) \cdot y_0^2)}$.

The programme uses a generalised form with the parameter $\alpha \in [0.5; 2]$. It transitions smoothly from the logarithmic law to the linear and square laws:

```
dA/dt = −a · A^(1−α/2) · B^(α/2) + r_A(t)
dB/dt = −b · B^(1−α/2) · A^(α/2) + r_B(t)
invariant without reinforcements:  b·A^α − a·B^α = const
```

2.3. Mathematical implications

The cost of numbers

By the square law, a side that is outnumbered twofold must exceed the adversary in quality (the product of lethality and survivability) fourfold. In general, a k-fold numerical shortfall requires quality k^α times greater. This refutes the intuition that “one modern vehicle is worth two old ones”: at α = 2 it must be worth four.

Principle of concentration

A force N divided into two equal parts, at α = 2, gives 2·(N/2)² = N²/2, that is, half as much. Hence the value of intelligence, communications and manoeuvre: they make it possible to concentrate forces and defeat the adversary piecemeal. In the index these means are assigned to the “multiplier” role (C4ISR, EW, AEW&C).

The “3 : 1” rule as a consequence of the model

For success, the attacker needs a force ratio R = (e*defence* / e*offence* · d)^(1/α), where d is the defence-advantage coefficient (terrain, fortifications, preparation). With equal quality of the sides:

Law exponent α	d = 1.5 (hasty defence)	d = 2.5 (prepared)	d = 3.0 (fortified area, city)
α = 1 (linear; default since 13.10.2026)	1.50 : 1	2.50 : 1	3.00 : 1
α = 1.5 (the methodology’s previous value)	1.31 : 1	1.84 : 1	2.08 : 1
α = 2 (quadratic)	1.22 : 1	1.58 : 1	1.73 : 1

The classical “3 : 1” rule corresponds to the linear law at d = 3 and is not a universal constant. If precision-guided weapons and drones make fire aimed (α grows), the defence advantage in terms of numbers shrinks. If, however, the defence digs in and disperses (d grows) and the battlefield is saturated with strike means, the required superiority grows. This is precisely the mechanism that explains the positional character of the war in Ukraine in 2022–2026 (section 4.1).

What the data showed (13.10.2026). According to the CDB90 database, with equal combat effectiveness of the sides the defence advantage in loss exchange is close to 1 (d ≈ 1.0–1.1 for hasty, prepared and fortified defence), and one step of superiority in effectiveness weighs more than a twofold numerical advantage. Therefore, in the methodology d = 1.3 for hasty defence and strikes and 1.0 for prepared and fortified; the d of a particular war can be measured from its loss ratio — for the war between Russia and Ukraine in 2024–2025, according to Oryx, d = 1.7 (document 6, sections 7.5–7.7). The table above remains an illustration of the formula.

2.4. Empirical validation

Battle and research	Result
Iwo Jima (Engel, 1954)	the quadratic law with reinforcements describes American troop losses well
Ardennes (Bracken, 1995)	best by residual sum of squares — the linear law
Ardennes (Fricker, 1998)	neither linear nor quadratic fits; on corrected data the “super-logarithmic” law performs better
Kursk (Clemens, 1997)	the logarithmic law performs better than the linear and quadratic ones
Kursk (Lucas, Turkes, 2004)	no basic law fits well and robustly; different laws give a comparable fit
Kursk (Lucas, Dinges), based on forces actually engaged	linear $R^2 = 0.62$; logarithmic 0.54; quadratic 0.29
Battle databases (Helmbold, Hartley)	the “Helmbold ratio” is linked to the initial force ratio; the data support intermediate laws
Battles of Solenopsis invicta ants	exponent ≈ 1 instead of 2: even in nature the square law does not hold
CDB90, 404 battles 1900–1973 (MPI pilot, 13.10.2026)	α = 0.65 [0.46; 0.84]; adjusted for combat effectiveness, 0.98 [0.80; 1.18], d ≈ 1; numbers hardly predict victory

Conclusion. The coefficients are non-stationary, and the “right” law depends on the level of combat, the type of weapon, the method of counting forces and the fitting method. Therefore α is a **calibration parameter**: since 13.10.2026, 1.0 by default (linear law, estimated on CDB90 battles and validated against the 10 wars of 2018–2025), in the uncertainty calculation — from 0.8 to 1.2; the previous value — 1.5. The model is useful for structural conclusions and comparison of scenarios, but not for an accurate forecast of losses. Validation against the 10 wars of 2018–2025 using data from the year before the war showed that the force ratio by the index carries almost no predictive information: model error σ = 2.0 in the logarithm of the ratio, so the programme shows the probability that the advantage is sufficient, and the outcomes of the battle model — as conditional (document 6, section 7.8).

2.5. Limitations when scaling to the state level

6. **Homogeneity.** Real forces are heterogeneous and must be reduced to a common unit (benchmark combat units).
7. **Constancy of coefficients.** In reality they change with the tempo of combat, fatigue, ammunition stocks and the adversary's adaptation.
8. **No space or logistics.** The model “mixes” forces at a single point, with no front, depth or supply.
9. **Command, intelligence, EW and morale are not accounted for.** These are multipliers, not addends.
10. **Nuclear deterrence is not added to conventional forces.** It is a threshold that limits the goals of war.
11. **No economy or time.** The outcome of a protracted war is decided by the defence industry, mobilisation and finances.
12. **No alliances** or their reliability.
13. **No geography.** Power decays with theatre distance (Boulding gradient).
14. **Asymmetric and hybrid conflicts are poorly described.**
15. **Closed data** on quality and readiness require expert assessments with explicit uncertainty.

3. Criticism of existing models

13 approaches are analysed below. For each, the essence, strengths, shortcomings with rationale and what we borrow are given. Evaluation criteria: transparency and reproducibility, allowance for quality and readiness, synergy of branches of service, the nuclear factor, geography, time and sustainment, allies, uncertainty, validation against real outcomes.

3.1. Global Firepower PowerIndex (GFP)

Essence. An annual ranking of 145 countries by “more than 60 factors” (per the 2026 descriptions). The final PwrIndx value is better the closer to zero (0.0000 — an unattainable ideal). Bonuses and penalties are applied after the base calculation. Nuclear weapons are not counted directly.

Strengths. Broad coverage, annual updates, clear presentation. This is the most cited rating in the media.

Criticism and rationale.

16. **Opacity and irreproducibility.** The formula, the weights and the composition of bonuses and penalties are not fully published. An independent researcher cannot reproduce the calculation, and therefore cannot check either the error or the sensitivity of the result to assumptions. A methodology that cannot be tested cannot serve as a basis for decisions.
17. **Quantity without quality.** Counting indicators (“tanks”, “aircraft”, “ships”) enter without adjustment for generation, serviceability and ammunition stock. Example: 1,000 tanks of obsolete generations against 300 modern ones. A linear count gives 3.3 : 1. If, however, a modern vehicle is 10 times superior to an old one in the product of lethality and survivability, then under the square law the forces are equal: $10 \cdot 300^2 = 900{,}000$ against $1 \cdot 1{,}000^2 = 1{,}000{,}000$. At $\alpha = 1.5$ the advantage is with the modern side: $10 \cdot 300^{1.5} \approx 52{,}000$ against $1{,}000^{1.5} \approx 31{,}600$, while at $\alpha = 1$, adopted in the methodology, — 3,000 against 1,000.
18. **No readiness.** Equipment in storage and unserviceable equipment are counted. The war of 2022–2026 has shown that the share of combat-capable equipment differs severalfold between armies.
19. **Rewarding diversity.** Emphasis on “diversity of armaments” leads to a paradox: a country scores points for having a class of equipment even if its combat value is small.
20. **Excluding nuclear weapons** removes the main factor limiting the goals of a war against a nuclear power. For the index this is a specification error: the factor cannot be excluded; it must be introduced correctly — as a threshold, not as a sum (section 5, U5).
21. **No uncertainty.** An exact rank is published (“4th place”), although differences between neighbouring countries are often smaller than the data error.
22. **Reality check.** In the GFP ranking of 2022 Russia stood in 2nd place, Ukraine — in 22nd. The counting gap suggested a rapid outcome, but the war became a protracted positional one: what proved decisive was the defence advantage, motivation, intelligence, drones and external support, which are not in the formula. In 2025 Israel and Iran stood close to each other in GFP (around 15th and 16th places). However, in June 2025 and in the 2026 campaign, Israeli aviation achieved air supremacy over Iran within the first days. The qualitative gap in the air force, air defence and intelligence was not seen by the counting index.

What we borrow. Coverage and presentation. We do not borrow the formula.

3.2. CINC (Correlates of War, National Material Capabilities)

Essence. Six components — military expenditure, armed forces numbers, pig iron and steel production, primary energy consumption, total and urban population. Each is expressed as a share of the world total, and the index is their equally weighted mean. Series are available from 1816.

Strengths. Full transparency, long series, the academic research standard.

Criticism and rationale.

23. **Obsolete proxies.** Steel and energy were a measure of industrial power in the era of the world wars. Today they poorly reflect the ability to produce missiles, electronics and drones.
24. **The double weight of population.** Total and urban population are two of the six components, that is, a third of the index. Therefore the index systematically overstates poor, populous countries. In recent editions China's CINC is noticeably higher than that of the United States, contrary to most expert assessments of the military balance.
25. **Equal weights** — an arbitrary choice not validated against outcomes.
26. **Shares of the world total** turn the index into a zero-sum game: growth of one country mechanically lowers the scores of the others.
27. **Weak predictive power.** Carroll and Kenkel (2019) showed that the CINC ratio of the sides predicts the outcomes of interstate disputes only slightly better than random guessing.
28. **No quality, readiness, geography, alliances or nuclear factor.**

What we borrow. The historical series for validation and the components as separate indicators (I21 — steel, L13 — energy, D12 — urban population) with small weights.

3.3. Lowy Institute Asia Power Index

Essence. 131 indicators in 8 measures: economic and military power, resilience, future resources, economic links, defence networks, diplomatic and cultural influence. Coverage — 27 countries of Asia. The military measure includes the armed forces, armaments and platforms, “signature” capabilities and military presence in the region.

Strengths. A transparent hierarchy with sub-measure weights, data publication, a strong interactive interface. This is the best presentation among comparable systems.

Criticism. (1) A mixing of hard and soft power: cultural influence in the same index as the armed forces complicates military interpretation. (2) A significant share of expert surveys. (3) Regional coverage. (4) No dynamics or scenarios. (5) The measure weights are expert-set and are not calibrated against outcomes.

What we borrow. The hierarchy “measure → sub-measure → indicator”, the concept of “signature capabilities” (hypersonics, aircraft carriers, SSBNs), and the presentation of the result.

3.4. BICC Global Militarization Index

Essence. It measures militarisation relative to society: military expenditure to GDP and to health spending, service members to population and to the number of physicians, heavy armament per capita. The indicators are log-transformed, normalised by min–max, weighted with subjective weights (5, 3, 4, 2, 2, 4) and brought to a 0–1000 scale.

Criticism. The index answers the question “how mobilised a state is”, not “how strong it is”. The high rank of Israel, Ukraine or Russia reflects the burden the war places on society, not power. The weights are expert-set.

What we borrow. The normalisation technique (log → min–max) and the relative burden indicators (E03, P04) — as indicators of sustainment, not of power.

3.5. Heritage Index of U.S. Military Strength

Essence. An annual assessment of the US armed forces along three dimensions — capacity, capability, readiness — on a five-level scale from “very weak” to “very strong”. Capacity is compared with a

benchmark. For example, in the 2022–2023 editions the army received a “weak” rating, as it stood at about 62% of the required strength.

Criticism. A single country is assessed. The scale is qualitative. The result depends on the adopted requirement (the “two-war” standard), so the assessment is normative (“how much there should be”) rather than comparative (“who is stronger”).

What we borrow. The “capacity — capability — readiness” triad as the structure of subpillars within the combat pillars.

3.6. Material Military Power (Souva, 2023)

Essence. Country-years from 1865 to 2019. Major naval, air and land armaments are counted, as well as nuclear weapons and ballistic missiles. According to the Modern War Institute, the measure correctly predicts the outcome of about 80% of the 36 wars in the COW catalogue.

Criticism. A small sample of wars. Platforms are counted without serviceability or ammunition stocks. There is no economy of replacement and no alliances. It is a step in the right direction (armaments matter more than “steel”), but not a complete model.

What we borrow. The focus on armaments and mandatory validation against war outcomes.

3.7. DOE score (Carroll, Kenkel, 2019)

Essence. An ensemble of machine-learning methods trained on the outcomes of interstate disputes (MID) using the CINC components. It is an order of magnitude more accurate than the simple ratio of capabilities.

Criticism. (1) A “black box”: the model does not explain why one side is stronger. (2) The inputs are limited to the CINC components. (3) Training on low-intensity disputes rather than on wars. (4) A change of technological era (drones, precision-guided weapons) is a distribution shift under which a model trained on the past loses accuracy.

What we borrow. Calibration of weights against historical outcomes and a benchmark ML model for testing our index.

3.8. “Net resources” (Beckley, 2018)

Essence. Power is measured net of the cost of maintaining the population. A rough proxy is GDP × GDP per capita. The approach corrects the overstatement of poor, populous countries.

Criticism. Wealth is potential, not mobilised power. In 2025 Russia spent 7.5% of GDP and 20% of state expenditure on defence (SIPRI), while many rich countries — 1–2% of GDP. The proxy does not account for the will and the ability to put the economy on a war footing.

What we borrow. Indicator E13 and the adjustment for “maintenance costs” in the “Defence economy” pillar.

3.9. Credit Suisse (2015)

Essence. A weighted sum of six categories: personnel, tanks, attack helicopters, aircraft, aircraft carriers and submarines. The weights increase towards the maritime component.

Criticism. Arbitrary weights, no quality or readiness, only six indicators. The result is extremely sensitive to the weight of aircraft carriers.

3.10. QJM and TNDM (T. Dupuy)

Essence. Combat power P = S × V × CEV. Here S is strength by weapon lethality indices (OLI), V — situational factors (terrain, weather, position, air supremacy), CEV — the combat effectiveness coefficient of the troops. The model was calibrated on the battles of the Second World War and the Arab–Israeli wars.

Criticism. Tactical level, a partly closed methodology, labour intensity. CEV was historically derived from the outcomes, which creates a risk of overfitting. Twentieth-century calibration does not account for drones or precision-guided weapons.

What we borrow. Explicit situational multipliers (the defence coefficient d, terrain) and the effectiveness coefficient as part of the “quality” role.

3.11. WEI/WUV and the potential–antipotential method

Essence. American methods for obtaining a single scale of heterogeneous armaments. WEI/WUV is a linear sum of models with expert weights. In the “potential–antipotential” method (TACWAR) the weights are found as the eigenvector of the kill-intensity matrix by the Perron–Frobenius theorem.

Criticism. Linearity does not account for synergy and competition among the elements of a force grouping. Expert coefficients are subjective. The eigenvector depends on the assumed distribution of fire. The estimates cannot be mixed with loss relationships derived on a different scale.

What we borrow. Benchmark coefficients for models (weapons and equipment reference) and weight calculation via the eigenvector. We add synergy through CES aggregation.

3.12. The domestic combat potential method

Essence. Heterogeneous means are reduced to a benchmark through commensurability coefficients. The force ratio is calculated from the sum of the combat potentials of formations, while the potentials of echelons are derived from the Osipov–Lanchester duel model and linear programming methods.

Criticism. The coefficients are closed and quickly become outdated as generations of weapons and equipment change, which is especially noticeable in the era of mass drones. The method belongs to the tactical–operational level and does not account for the economy or the resilience of the state. Potentials are summed linearly.

What we borrow. Benchmark combat units (CP) and commensurability coefficients. We supplement them with an annual review of the reference and non-linear aggregation.

3.13. The Lanchester model itself as an index

Using the Lanchester equations directly as a rating is also mistaken. The model is deterministic, homogeneous, takes no account of space, logistics, economics or alliances, and its laws are poorly confirmed by history (section 2.4). Therefore in the programme Lanchester is a **form** (the weight rule, the link between quality and numbers) and a **scenario engine**, but not the sole measure of power.

3.14. Comparative assessment of approaches

Legend: “+” — accounted for, “±” — partially, “–” — not accounted for.

Model	Transparency	Quality	Readiness	Synergy	Nuclear	Geography	Time	Alliances	Uncertainty	Validity
GFP	–	±	–	–	–	±	±	–	–	–
CINC	+	–	–	–	–	–	±	–	–	±

Model	Transparency	Quality	Readiness	Synergy	Nuclear	Geography	Time	Alliances	Uncertainty	Validity
Lowy API	+	±	±	−	±	±	±	+	−	−
BICC GMI	+	−	−	−	−	−	−	−	−	−
Heritage	±	+	+	−	±	±	−	±	−	−
Souva MMP	+	±	−	−	+	−	−	−	−	+
DOE	±	−	−	±	−	−	−	−	±	+
Beckley	+	−	−	−	−	−	±	−	−	+
QJM/TNDM	±	+	±	±	−	+	−	−	−	+
WEI/WUV	±	+	−	−	−	−	−	−	−	±
Combat potentials	−	+	±	−	−	±	−	−	−	±
MPI-LO	+	+	+	+	+	+	+	+	+	+

4. Lessons of the 2022–2026 wars for the methodology

The recent wars have provided a rare data set for testing methodologies. Each lesson below has been turned directly into an element of the programme.

4.1. Russia — Ukraine: the dominance of defence and the drone economy

By October 2026 the war is in its fifth year. According to open-source estimates, the front line has shifted slowly over the years, while the sides wage an intense war of long-range strikes on infrastructure. On 11 October 2026 an “energy truce” was announced, but the strikes continued.

- **The level of d has grown.** The saturation of the battlefield with reconnaissance and strike drones has made any concentration visible and vulnerable. According to the formula in section 2.3, a growth of d at $\alpha \approx 1.2–1.5$ raises the ratio required for an offensive to 2–3 : 1. Such a ratio is unattainable along a broad front, and the model predicts a positional war of attrition, which is what is observed.
- **Production matters more than stocks.** In 2025 Ukraine reportedly produced more than 4 million drones and plans 7–10 million in 2026. Russia is ramping up the output of shells, glide bombs and strike drones. Therefore the catalogue includes annual output indicators (G22, I06–I10) and the “sustainment” role.
- **Manpower is a constraint.** In 2026 Russia is estimated to recruit about 30 thousand people per month, with a downward trend. Ukraine is experiencing mobilisation difficulties. The reserve R and the replacement rate r(t) are therefore introduced into the model.
- **External support is a separate channel.** Allied assistance (the EU, the coalition of the willing, purchases) affects primarily sustainment, not numbers. The coalition module therefore distinguishes between types of contribution.

4.2. Israel — Iran (2025–2026): quality and the “weakest link”

In June 2025 and in the campaign of the United States and Israel from 28 February 2026 (a truce on 8 April, the resumption of fighting from 8 July, a US naval blockade), Israeli and American aviation quickly suppressed Iran's air defence. Iran's supreme leader Ali Khamenei was killed; Mojtaba Khamenei was appointed as his successor.

- **The weakest link decides.** Iran has large ground forces and a powerful missile arsenal, but a weak air force and air defence opened the country to strikes throughout its depth. A linear sum of the pillars would have “covered” this weakness with the strength of other components. CES aggregation with $\rho < 1$ does not allow this (U4).
- **Survivability is part of quality.** According to US intelligence (May 2026, the estimate is disputed), Iran retained about 70% of its pre-war missile stock and launchers thanks to its underground infrastructure. Therefore protection and shelters are included in the “quality” role, and survivability in the battle model.
- **Asymmetric maritime denial.** The threat in the Strait of Hormuz (mines, boats, anti-ship missiles) affects the world economy out of proportion to the size of Iran's navy. This is captured by indicators D09 and N09 and by the salvo model.
- **Multipliers matter more than mass.** Intelligence, EW, AEW&C, tankers and precision-guided means were decisive. These are roles M and P.

4.3. US operation in Venezuela (January 2026): projection and C4ISR

On 3 January 2026 the United States conducted Operation “Absolute Resolve” with strikes on air defence and the capture of Nicolás Maduro. With a huge gap in power projection, intelligence and command, the

outcome proved instantaneous. Hence the conclusion: with a large force ratio and multipliers present, what matters is not the numbers of the victim's armed forces, but the projection radius λ and the role M of the attacker.

4.4. The Houthis and the Red Sea (2023–2025): the economics of exchange

Cheap drones and missiles against expensive interceptors have created a cost asymmetry: the defender spends more than the attacker. The programme therefore includes the Hughes salvo model and a cost-exchange indicator (cost of interception to cost of attack).

4.5. India — Pakistan (May 2025) and the nuclear threshold

A short conflict between two nuclear powers stayed below the nuclear threshold. Nuclear weapons limit the scale and goals of a war, but do not exclude conflict: this is the “stability–instability paradox” (Snyder, 1965). Therefore the deterrence axis in the MPI limits the achievable goals of a scenario, rather than simply adding points (U5).

4.6. Local wars of 2025–2026

Pakistan and Afghanistan have been waging an open armed conflict since February 2026, while Thailand and Cambodia — border fighting in 2025. In both cases the local superiority of one side is limited by political goals, terrain and external mediation. The methodology accounts for this through the terrain coefficient, the break threshold and the scenario's goal constraints.

5. Improvements built into the programme

Each improvement is described using the scheme “problem → solution → rationale → formula”.

U1. Roles of indicators

Problem. In comparable systems all indicators are “added up” in the same way, although numbers, quality, multipliers and resources affect the outcome differently.

Solution. Each of the 253 indicators has been assigned a role: N — numbers (41), Q — quality (57), M — multiplier (35), S — sustainment (81), P — projection (13), G — deterrence (11), C — context (15).

Rationale. In the combat equations, numbers stand in the exponent α, quality and multipliers in the effectiveness coefficient, sustainment in the reinforcement term, and projection in the decay with distance. The role determines where an indicator enters the formula, and thereby eliminates the mixing of heterogeneous quantities.

U2. Lanchester law within pillars

Problem. A weighted mean does not reflect the fact that numbers and quality interact multiplicatively.

Solution. In the combat pillars (ground forces, air force and air defence, navy) the weight of the “numbers” group is multiplied by α:

```
P_d = (α·Σw·x̃[N] + Σw·x̃[Q,M] + Σw·x̃[other]) / (α·Σw[N] + Σw[Q,M] + Σw[other])
```

Rationale. Counting quantities are normalised on a logarithmic scale. Then ln F = ln Q + α·ln N becomes a linear combination of normalised logarithms with the coefficient α on numbers. At α = 1 the formula reduces to an ordinary mean, and at α = 2 numbers weigh twice as much, as the square law requires.

U3. Benchmark combat units (CP)

Problem. “A tank is a tank” — comparable systems make no allowance for generation or serviceability.

Solution. N*eff* = $\Sigma_k\, n_k \cdot q_k \cdot r_k$, *where q is the quality coefficient relative to the benchmark (= 1.0), r is the share of serviceable units. The coefficients q are set by expert judgement using the analytic hierarchy process (AHP) across groups of performance characteristics and refined by the “potential–antipotential” method:* $v = (1/\lambda\max)\cdot K\cdot v$.

Rationale. Combining the domestic practice of benchmarking with the American eigenvector method yields both expert meaningfulness and internal consistency (the weight of a model reflects what it kills and with what intensity).

U4. Non-linear CES aggregation and the “weakest link”

Problem. With a linear sum, a shortage of air defence is offset by a surplus of tanks. Iran in 2026 showed that such offsetting does not work in reality.

Solution.

```
MPI = ( Σ_p ω_p · P_p^ρ )^(1/ρ)        ρ = 1 — sum;  ρ → 0 — geometric mean;  ρ < 0 — “weakest link”
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Rationale. The CES function is the standard in economics for describing complementary factors of production. Branches of service are exactly such factors: without air defence, aviation perishes at its airfields; without aviation and air defence, the ground forces do. By default ρ = 0.5 (partial substitutability), under calibration — 0.3–0.7. Sensitivity to ρ is shown in the interface.

U5. The nuclear axis and three regimes

Problem. Comparable systems either add warheads to tanks (meaningless offsetting) or exclude nuclear weapons (a specification error).

Solution. A separate deterrence axis: class 0–4 (no nuclear / threshold / limited / two components / a triad with assured retaliation) and the STR pillar score. The user chooses the integration mode: “separate axis” (by default), “as a pillar” or “multiplier” $(1 + \kappa \cdot P_STR)/(1 + \kappa)$.

Rationale. The marginal value of the thousandth warhead for deterrence is close to zero, while the existence of assured retaliation changes the adversary's admissible targets. Deterrence therefore sets **scenario constraints** (one cannot set the goal of destroying a state that has a triad), not scores.

U6. Power projection

Problem. In the ratings, a country on another continent “threatens” just like a neighbour.

Solution. $MPI(D) = MPI \cdot e^{(-D/\lambda)}$, where $\lambda = \lambda_0 + \lambda_1 \cdot \bar{P}[\text{role P}]$. The distance D is measured from the nearest major bases, including forward ones.

Rationale. This is Boulding's loss-of-strength gradient (1962). Aerial tankers, military airlift, aircraft carriers and overseas bases increase λ, that is, the distance over which a force loses 63% of its effectiveness.

U7. Time and sustainment

Problem. A static index does not show who will withstand a long war.

Solution. Indicators with role S provide the reserve R, the replacement rate r(t) and the mobilisation delay. The “Protracted war” preset raises the weights of the economy, the defence industry and logistics. The scenario horizon is chosen from 30 to 720 days.

Rationale. The war of 2022–2026 is a war of production: the monthly output of shells, drones, missiles and interceptors has proved more important than pre-war stocks.

U8. Alliances as explicit coalitions

Problem. In comparable systems alliances are either absent or give “points for membership” regardless of whether the ally will enter the war.

Solution. Coalition module: $F_{coal} = F_{own} + \Sigma_j \rho_j \cdot s_j \cdot F_j \cdot e^{(-D_j/\lambda_j)} \cdot g(t - \tau_j)$. Here ρ is reliability (the probability of entry), s is the share of forces allocated to the theatre, τ is the entry delay, g is the activation function. The types of contribution differ: forces, supply, intelligence, territory, finances. In a coalition scenario the “Alliances” pillar is excluded so that allies are not counted twice.

Rationale. Ally reliability has been tested by the wars of 2022–2026: North Korea sent troops to Russia, whereas Russia did not intervene in the war of Israel and the United States with Iran, despite the 2025 strategic partnership treaty. Details — in document 3.

U9. Uncertainty

Solution. Each value receives an Admiralty code (source reliability A–F × information reliability 1–6), which is translated into σ. Missing values are imputed with a flag and an increased σ. The Monte Carlo method varies values, weights (±15%), α and ρ and produces the MPI interval (5–95%) and a range of ranks.

Rationale. Publishing an exact rank when the error is comparable to the gap between countries is misleading. In the assessment of eight countries (document 2), ranks 2 to 6 are statistically indistinguishable, and an honest methodology is obliged to show this.

U10. Weight calibration

Solution. Three sources of weights: expert (AHP with consistency checking CR < 0.1 and Delphi), statistical (the entropy method, PCA for detecting duplicates) and calibration (fitting ω, α, ρ against the outcomes of wars and disputes from COW, MID and Souva with minimisation of the Brier score). The final weights are their compromise. Sensitivity is checked with Sobol indices.

U11. Scenario engine

Solution. The generalised Lanchester–Helmbold model with reinforcements is the main one. In addition: a heterogeneous model by “land–air–sea” domains, a salvo model for missile and naval strikes, stochasticity (Monte Carlo over inputs and loss noise), coalitions. The output — probabilities of three outcomes (offensive success, protracted fighting, defence success), time to break, losses, required and actual force ratio, sensitivity to α.

U12. Transparency and reproducibility

Solution. An open indicator catalogue, formulas in the interface with numbers substituted, storage of the methodology version and the data snapshot hash for each calculation, and a change log of expert assessments with justifications.

U13. Indicators of the drone era

Solution. The catalogue has been supplemented with indicators that comparable systems lack: the annual output of FPV drones and loitering munitions (G22), long-range strike drones (A13), counter-UAV means (G18), unmanned surface vessels (N17), UAV output (I10), as well as cost-exchange indicators in the scenario module.

6. Summary of formulas

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1. Transformation:   x̃ = log(1+x) for count and monetary data; share; score 0–10;
inversion for "↓"
2. Normalisation:       a global scale with reference-year anchors (main) or min–max
3. CP:              N_eff = Σ n_k·q_k·r_k;   q — AHP + potential–antipotential
4. Combat pillar:     P = (α·Σw·x̃[N] + Σw·x̃[Q,M] + Σw·x̃[O]) / (α·Σw[N] + Σw[Q,M] +
Σw[O])
5. Aggregation:          MPI = (Σ ω·P^ρ)^(1/ρ),  ρ = 0.5 (0.3–0.7)
6. Deterrence:      class 0–4; modes: axis | pillar | multiplier (1+0.3·P_STR)/1.3
7. Projection:         MPI(D) = MPI·e^(–D/λ),  λ = λ₀ + λ₁·P̄[P]
8. Coalition:         F = F_own + Σ ρ_j·s_j·F_j·e^(–D_j/λ_j)·g(t–τ_j)
9. Battle model:       dA/dt = –a·A^(1–α/2)·B^(α/2) + r_A(t);  a = k·e_B,  b = k·e_A/d
10. Required ratio:  R = (e_B·d/e_A)^(1/α)
11. Uncertainty: Monte Carlo over x ~ N(x̃, σ²), ω·LogN(0; 0.15²), α ~ U[1.2; 1.8], ρ ~
U[0.3; 0.7]
```

7. Validation

29. **Retrospective on wars.** Forecasting the outcomes of interstate wars and disputes of 1946–2024 from data of the year the conflict began; AUC metrics and the Brier score; comparison with CINC, GFP, DOE and MMP. The goal is to be no worse than DOE and MMP.
30. **Battle model on battles.** Fitting α and the coefficients on historical series (Iwo Jima, the Ardennes, Kursk, battle databases) with validation on a held-out sample.
31. **Validation against the wars of 2022–2026.** The model must reproduce the positional character of the war in Ukraine, the rapid outcome of operations with a large force ratio (Venezuela) and the suppression of air defence with a “weakest link” (Iran).
32. **Expert validation.** Pairwise comparisons of countries by experts → the Bradley–Terry model → agreement with the rating (Kendall's $\tau \geq 0.7$).
33. **Stability over time.** Every change of rank over a year must be explained by specific indicators.

8. Limitations and correct use

The MPI assesses potential, not intentions, resolve or the outcome of a particular war. Political will, strategy, surprise, the quality of leadership in a crisis and external support are formalised only partially. The results are intended for analytics, training and comparison of scenarios. They cannot be used as the sole basis for decisions, and conclusions are always accompanied by intervals, the methodology version and the data cut-off date.

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